WNet: W-Shaped Hierarchical Network for Remote-Sensing Image Change Detection

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***Abstract*— Change detection (CD) is a hot research topic in the remote-sensing (RS) community. With the increasing availability of high-resolution (HR) RS images, there is a growing demand for CD models with high detection accuracy and generalization ability. In other words, the CD models are expected to work well for various HRRS images. Convolutional neural networks (CNNs) have been dominated in HRRS image CD due to their excellent information extraction and nonlinear fitting capabilities. However, they are not skilled in modeling long-range contexts hidden in HRRS images, which limits their performance in CD tasks more or less. Recently, the Transformer, which is good at extracting global context dependencies, has become popular in the RS community. Nevertheless, detailed local knowledge receives insufficient emphasis in common Transformers. Considering the above discussion, we combine CNNs and Transformers and propose a new W-shaped dual-Siamese branch hierarchical network for HRRS image CD named W-shaped hierarchical network (WNet). WNet first incorporates a Siamese CNN and a Siamese Transformer into a dual-branch encoder to extract multilevel local fine-grained features and global long-range contextual dependencies. Also, we introduce deformable ideas into the Siamese CNN and Transformer to make WNet understand the critical and irregular areas within HRRS images. Second, the difference enhancement module (DEM) is developed and embedded into the encoder to produce the difference feature maps at different levels. Using simple pixel-wise subtraction and channel-wise concatenation, the changes of interest and irrelevant changes can be highlighted and suppressed in a learnable manner. Next, the multilevel difference feature maps are fused stage by stage by CNN–Transformer fusion modules (CTFMs), which are the basic units of the decoder in WNet. In CTFMs, the local, global, and cross-scale clues are taken into account to ensure the integrity of information. Finally, a simple classifier is constructed and added at the top of the decoder to predict the change maps. Positive experimental results counted on four public datasets demonstrate that the proposed WNet is helpful in HRRS image CD tasks. Our source codes are available at https://github.com/TangXu-Group/Remote-Sensing-Image-Change-Detection/tree/main/WNet.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), Transformer.**

I. Introduction

CHANGE detection (CD) is one of the significant topics in remote-sensing (RS) image processing, which aims at detecting changes of interest in bitemporal images covering the same regions [1]. It can be used in various RS applications, such as land cover recognition [2], urban expansion estimation [3], disaster assessment [4], and agriculture monitoring [5]. With the development of sensors, many high-resolution (HR) RS images can be generated daily. These images bring profitable information and formidable challenges for CD tasks at the same time [6]. For example, as shown in Fig. 1, due to various environmental conditions, the inevitable irrelevant changes would disturb the CD results negatively [7]. Furthermore, as illustrated in Fig. 2, resolutions of HRRS images are diverse and varied, which places higher demands on the generalization ability of the CD model [8]. In other words, it is difficult for a model to perform well in different HRRS images with various scales. Therefore, developing specific CD methods for HRRS images is necessary and urgent.

Nowadays, with the rapid development of deep learning [9], [10], [11], [12], convolutional neural networks (CNNs) have dominated the HRRS image CD community. Benefiting from the strong capabilities of feature learning and nonlinear fitting, many CNN-based CD methods have been proposed for HRRS images and achieved success in their applications [13], [14], [15]. Even so, the researchers found that CNNs are not good at exploring the context information and discovering the salient areas within HRRS images. To mitigate these problems and improve the CD results, some works stack more convolution layers [16], [17] and others embed advanced techniques (such as visual attention mechanisms [18], [19]) into CNNs [20], [21], [22]. Although they perform well, these approaches are not proficient in modeling the long-range context dependencies hidden in HRRS images, which are important clues for CD tasks.

The current Transformer [23], especially the visual Transformer (ViT) [24], has effectively solved the shortcomings of CNNs mentioned above. With the excellent ability to

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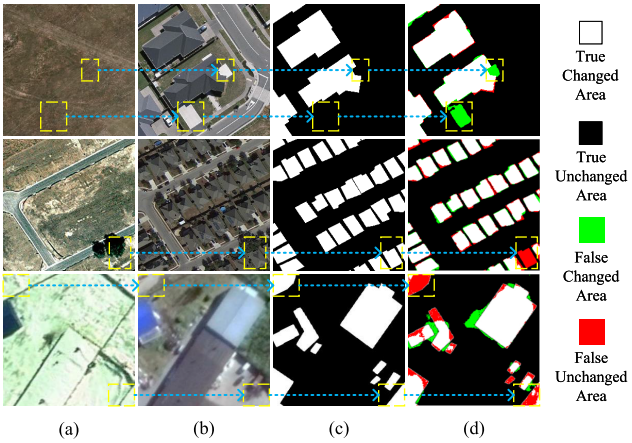


Fig. 1. Three irrelevant changes in three pairs of bitemporal HRRS images are caused by nonbuildings changes in building CD (the first row), illumination condition change (the second row), and seasonal change (the third row), respectively. (a) T1 image. (b) T2 image. (c) Ground truth. (d) Possible change prediction map.

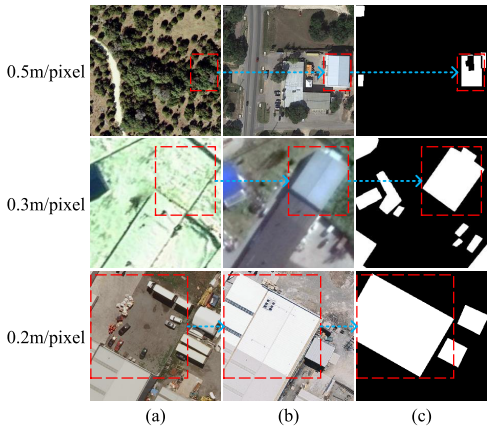


Fig. 2. Resolutions of the three sets of temporal HRRS images are 0.5, 0.3, and 0.2 m/pixel from top to bottom. Although the changed objects in them are both buildings, their scales are distinctly varied, which requires the CD model to have a high generalization ability. (a) T1 image. (b) T2 image. (c) Ground truth.

explore long-range context dependencies, the Transformer has been gradually applied to HRRS image CD tasks. Sufficient and positive experimental results confirm its effectiveness in HRRS image CD tasks [25], [26], [27], [28]. While the Transformer offers great potential in CD, there is still room for improvement. First, although the global contexts hidden in HRRS images can be captured by the Transformer, the local fine-grained messages do not get the attention they deserve. This would impact the CD results negatively. Second, most Transformer-based models would divide input HRRS images into patches using the simple grid scheme for downstream operations. However, these regular patches would disturb the continuity and geometry of complex land covers, which goes against the CD's targets.

Based on the above discussion, we found that CNNs and Transformers are the two complementary tools for HRRS image CD. Therefore, we propose a W-shaped hierarchical network (WNet) in this article to deal with HRRS image CD tasks by integrating the superiorities of CNNs and Transformers. First, WNet consists of parallel CNN and Transformer branches. They aim to learn the fine-grained local messages and long-range context dependencies from HRRS images. Using those features simultaneously is beneficial to identify the changing objects with various scales. Furthermore, we embed the deformable convolution [29], [30] and deformable attention mechanism [31] into CNN and Transformer branches to weaken the adverse effects of fixed convolution kernels and regular patches. Third, a difference enhancement module (DEM) is developed to emphasize the change information. This way, the irrelevant changes caused by the environment can be discovered, which is good for reducing false alarm rates. Finally, we design a CNN–Transformer fusion module (CTFM) based on an improved channel attention strategy [18]. CTFM focuses on enriching the channel clues and integrating the local, global, and cross-scale features. As a whole, the dual-Siamese branch structure of WNet ensures that it can obtain suitable feature representations to detect the change information hidden in bitemporal RS images collected under various environmental conditions. The introduction of deformable technologies improves WNet's generalization ability so that it works for different HRRS images.

The main contributions of our work can be summarized as follows.

1) An end-to-end HRRS image CD model named WNet is developed in this article. WNet is a hierarchical network with a dual-Siamese branch encoder, which involves a Siamese CNN and a Siamese Transformer. It aims to capture local and global information from bitemporal HRRS images simultaneously.
2) To enrich the WNet's receptive fields, we introduce the deformable ideas into the dual-Siamese branch encoder. In this way, the limitations caused by the fixed convolutional kernel (in CNNs) and regular patch generation (in Transformers) can be mitigated.
3) To highlight the changes of interest and suppress irrelevant changes, we develop DEM and embed it into the dual-Siamese branch encoder. By combining simple subtraction and concatenation operations, the difference feature maps can be obtained in an efficient and learnable way.
4) To fuse diverse information, the CTFM is developed. In CTFM, not only can the local, global, and cross-scale messages be integrated, but also the channel-wise clues are mined intensely. In this fashion, the integrity of final features for change map generation can be guaranteed.

The rest of this article is organized as follows. The literature related to CNN- and Transformer-based methods for RS CD is reviewed in Section II. Then, the proposed method is introduced in Section III. The experimental results and discussion are given in Section IV. Finally, the conclusions are summarized in Section V.

II. Related Work

A. CNN-Based Methods

The CNN has become a standard technique in the field of HRRS image CD. Many CNN-based models have been proposed to address HRRS image CD tasks in recent years. In the

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beginning, researchers are used to introducing some classical CNNs (such as VGG [32] and ResNet [33]) to be the feature extractor for the final change maps [34], [35], [36]. Then, some semantic segmentation networks are used to deal with HRRS image CD since the CD task can be regarded as a two-class classification problem. For instance, Duadt et al. [37] designed three CD networks for HRRS images based on UNet [38]. Three models are named FC-EF, FC-Siam-conc, and FC-Siam-diff according to ways of input and fusion. The positive results confirm that UNet and full convolutions are useful to HRRS image CD. Another example of the UNet-based model was proposed in [39]. Besides adopting UNet, a feature fusion strategy is developed to fuse the multilevel features. In this fashion, the multiscale information within HRRS images can be mined to ensure the quality of change maps. As the study progressed, the researchers found that the Siamese architecture is more suitable for CD than the single-branch structure. Thus, a growing number of HRRS image CD networks with the Siamese style have been presented [40], [41], [42].

Although the methods mentioned above have succeeded, the shortcomings of CNNs still limit their performance, that is, the fixed convolutional kernels are not able to capture the contextual information and emphasize crucial areas within HRRS images for CD. To address these issues, many studies [43], [44], [45] have incorporated visual attention mechanisms [46] and altered advanced strategies to improve the performance of CNN-based CD networks. Fang et al. [43] proposed a densely connected Siamese network named SNUNet, in which an improved channel attention module (CAM) [18] is adopted to discover the context information and enhance the discrimination of learned features. The compact information transmission and adequate information representation enable SNUNet to demonstrate excellent CD performance. Similarly, a dual-branch multilevel intertemporal network was introduced in [45], where a joint attention module is designed to exchange bitemporal feature information for further generating difference features. Based on these features, favorable change maps can be obtained by subtraction or concatenation. Apart from including the attention mechanisms, some other studies focus on enhancing or utilizing the boundaries of changed objects to improve the CD performance of networks [47], [48], [49].

B. Transformer-Based Methods

Recently, ViT [50], [51], [52], [53], [54] has been increasingly applied to HRRS image CD due to its specific characteristic, that is, the strong capacity to model the global long-distance relations hidden in images. The current Transformer-based CD methods can be broadly classified into two categories. The methods within the first category embed the existing Transformers into their models for discovering the changes [55], [56], [57]. For example, Bandara and Patel [55] proposed a ChangeFormer. It first uses the pyramid ViT (PVT) [58] as the basic unit to construct a Siamese encoder to learn valuable features from bitemporal HRRS images. Then, a multilayer perception (MLP) decoder is developed to complete the feature fusion and change map prediction. The encouraging results illustrate the effectiveness of ChangeFormer. Likewise, a U-shaped network (SwinSUNet) was constructed in [56] based on the Swin Transformer (Swin-T) [59]. SwinSUNet consists of an encoder, a fusion module, and a decoder. Their units are all Swin-T blocks. In addition, the skip connection strategy is added between the encoder and the decoder, which further enhances the learned features. Combining them, encouraging HRRS image CD results can be generated with acceptable time costs. Although these approaches demonstrate the potential of the Transformer in HRRS image CD, the missing local detailed information impacts their behavior negatively.

To mitigate the above problems, the models within the second category pay attention to integrating the advantages of CNNs and Transformers for HRRS image CD tasks [26], [60], [61], [62], [63]. Chen et al. [60] proposed a bitemporal image Transformer (BIT) model for HRRS image CD, in which the CNN and Transformer are linked in series. BIT first leverages ResNet [33] to map bitemporal HRRS images into different feature maps. Then, the original Transformer block [24] is adopted to build the spatial–temporal context within the features. The satisfying CD results imply the success of the combination of CNNs and Transformers. Based on BIT, another hybrid network was presented in [26] and named TransUNetCD. Like BIT, the CNN and Transformer are constructed serially to explore the local and global information from HRRS images. Unlike BIT, TransUNetCD embeds this combination into a UNet to improve performance. In contrast to directly applying Transformers to process deep features extracted by CNNs, Feng et al. [61] designed a dual-branch network named ICIF-Net, where CNNs and Transformers are placed in each branch. When bitemporal HRRS images are input ICIF-Net, the CNN and Transformer blocks would process them, respectively. Then, the CNN- and Transformer-based features are fused crosswise from intra- and interscale aspects. Consequently, the local detailed and global context knowledge hidden in HRRS images can be fully explored for the final CD.

III. Methodology

A. Overall Framework

The overall framework of our WNet is shown in Fig. 3, which consists of a dual-branch encoder and a fusion decoder. When we input two bitemporal HRRS images $\mathbf{I}_1 \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{I}_2 \in \mathbb{R}^{H\times W\times C}$ into WNet, they will be passed through the dual-branch encoder first, which encloses a Siamese CNN branch and a Siamese Transformer branch. Thus, the multilevel local fine-grained information $\{(\mathbf{F}_i^1, \mathbf{F}_i^2), i = 1, 2, 3, 4\}$ and the global context dependencies $\{(\mathbf{G}_i^1, \mathbf{G}_i^2), i = 1, 2, 3, 4\}$ can be generated. Meanwhile, the embedded DEMs are used to produce the difference feature maps $\{\tilde{\mathbf{F}}_i, i = 1, 2, 3, 4\}$ and $\{\tilde{\mathbf{G}}_i, i = 1, 2, 3, 4\}$, in which the change information of bitemporal HRRS images can be highlighted efficiently. Then, the obtained difference feature maps are fed into the fusion decoder, which is made up of CFTMs and a classifier, to aggregate the various knowledge from CNN and Transformer branches and to generate the change map $\mathbf{M} \in \mathbb{R}^{H\times W}$.

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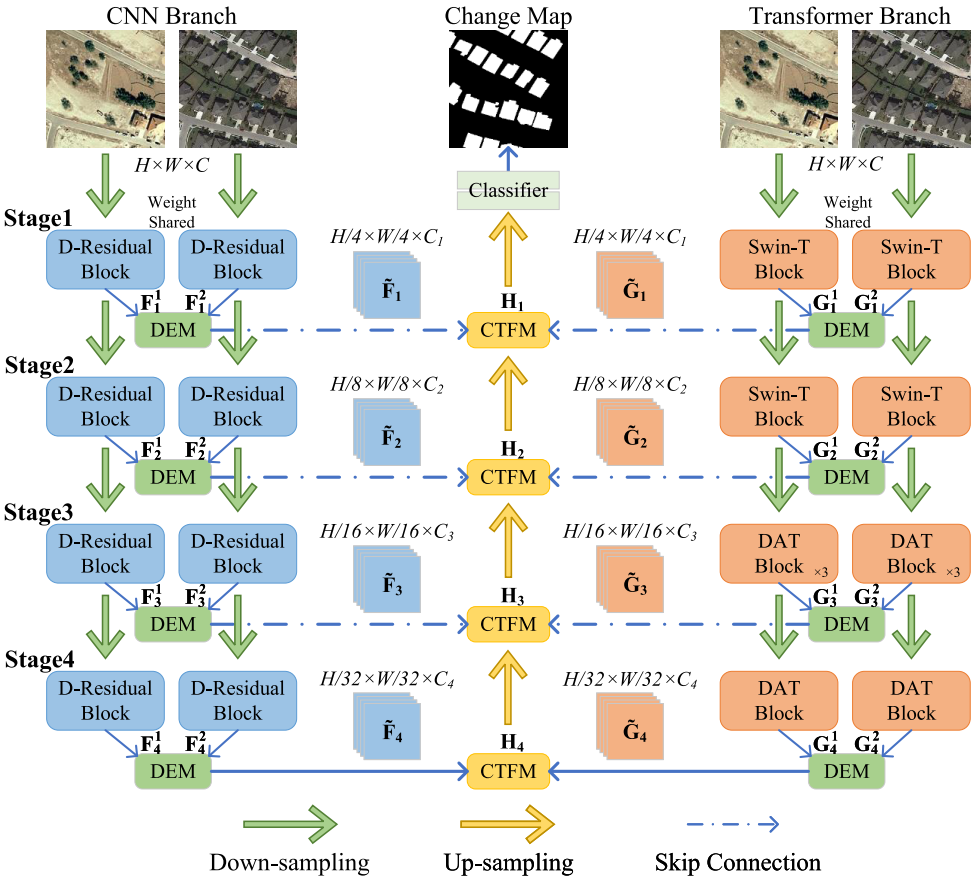


Fig. 3. Framework of the proposed WNet. Multilevel features are extracted from bitemporal HRRS image pairs through Siamese CNN and Siamese Transformer branches. Then, the difference feature maps are generated by DEMs at different stages. After that, the local, global, and cross-stage information is fused using CTFMs stage by stage. Finally, a simple classifier is designed to generate change maps.

In addition, skip connections are applied to connect the encoder and the decoder at different levels to ensure the stability of WNet.

B. Dual-Siamese Branch Encoder

1) Difference Enhancement Module: The main components of the dual-branch encoder are a Siamese CNN and a Siamese Transformer. Both of them leverage DEM to generate difference feature maps at various levels. Thus, we first explain DEM in this section. The structure of DEM is exhibited in Fig. 4. Assume that the input data is two feature maps $\mathbf{Z}_1 \in \mathbb{R}^{h\times w\times c}$ and $\mathbf{Z}_2 \in \mathbb{R}^{h\times w\times c}$ corresponding to the bitemporal feature maps. DEM first uses pixel-wise subtraction to obtain the difference feature map $\mathbf{Z}_{\text{Diff}} \in \mathbb{R}^{h\times w\times c}$. Then, the original feature maps are concatenated with the difference feature map at the channel-wise to get $\mathbf{Z}_{\text{EN}} \in \mathbb{R}^{h\times w\times 3c}$, in which the change information can be highlighted. Finally, a 1×1 convolution layer is applied on $\mathbf{Z}_{\text{EN}}$ to further learn change information, reduce the channel dimension, and produce the output $\mathbf{Z}_{\text{DEM}} \in \mathbb{R}^{h\times w\times c}$. The above process can be formulated as

$$\begin{aligned}\mathbf{Z}_{\text{DEM}} &= f_{\text{Conv}1\times 1}(\mathbf{Z}_{\text{EN}})\\ \mathbf{Z}_{\text{EN}} &= \text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \quad \mathbf{Z}_{\text{Diff}}), \mathbf{Z}_{\text{Diff}} = |\mathbf{Z}_1 - \mathbf{Z}_2| \end{aligned} \tag{1}$$

where $|\cdot|$, Concat$(\cdot)$, and $f_{\text{Conv}1\times 1}(\cdot)$ denote the absolute value, channel-wise concatenation, and 1×1 convolution layer, respectively.

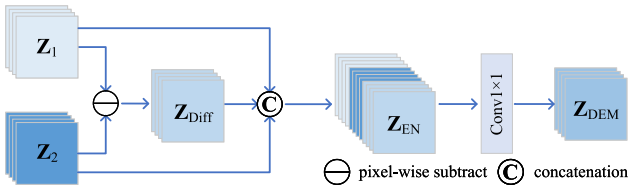


Fig. 4. Structure of DEM, which mainly contains pixel-wise subtract, concatenation, and 1×1 convolution.

Here, pixel-wise subtraction can highlight highly divergent changes in bitemporal feature maps, including changes of interest and irrelevant changes. The channel-wise concatenation aims to add prior knowledge of change information to bitemporal feature maps, which enhances the representation of change information. Along with the subsequent 1×1 convolution and CTFM in the decoder, the changes of interest can be gradually approximated, and the irrelevant changes can be depressed.

2) Siamese CNN Branch: The backbone of our Siamese CNN branch is ResNet18 [33]. It has four residual blocks, each with two basic residual units (see Fig. 5). Every unit has two 3×3 convolutions, two batch normalization (BN) layers, and two rectified linear units (ReLU) activation functions, which are connected in a residual way. Although the residual blocks have successed in many applications, their performance in HRRS image CD is also limited because the fixed and regular

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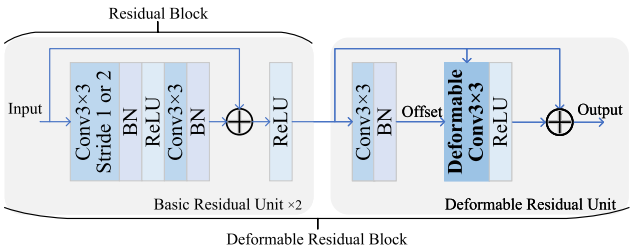


Fig. 5. Architecture of the D-Residual block, which consists of a residual block and a deformable residual unit. The residual block consists of two basic residual units.

convolution kernels decrease their ability to explore the geometric characteristics of changed/unchanged areas. To mitigate this issue, we introduce the deformable convolution [30] and develop the deformable residual (D-Residual) block in this section.

The key idea of the deformable convolution is to add the direction clues into the convolutional kernels so that the receptive field can be enriched. For an input feature map $\mathbf{x}$, the process of a normal convolution can be defined as

$$\mathbf{y}(p) = \sum_{p_n \in \mathcal{R}} \mathbf{w}(p_n) \cdot \mathbf{x}(p_n) \tag{2}$$

where $\mathcal{R}$ implies a regular grid, p represents a position in $\mathcal{R}$, and $\mathbf{w}$ denotes the convolutional kernel. To augment the receptive field, the deformable convolution upgrades the regular grid $\mathcal{R}$ by including a learnable offset Δp_n. Therefore, the deformable convolution process can be formulated as

$$\mathbf{y}(p) = \sum_{p_n \in \mathcal{R}} \mathbf{w}(p_n) \cdot \mathbf{x}(p_n + \Delta p_n). \tag{3}$$

This way, the receptive field can be enlarged to an irregular range controlled by the offset locations $p_n + \Delta p_n$. More details of the deformable convolution can be found in [29] and [30].

The structure of the D-Residual block is shown in Fig. 5. It contains a residual block, a 3×3 convolution, a 3×3 deformable convolution, a BN layer, and a ReLU activation. Also, the residual block and other parts are connected in a skipped manner. Based on the D-Residual block, our Siamese CNN branch is developed hierarchically. Particularly, this branch has two identical CNNs, each of which has four stages embedded D-Residual blocks. Like ResNet18, a 7×7 convolution and a max pooling layer are added before Stage 1 for downsampling. The downsampling process in other stages is achieved by setting the stride of the first 3×3 convolution in the first basic residual unit to 2. Furthermore, two CNNs are connected by DEMs at different stages. Consequently, after passing the bitemporal HRRS images $\mathbf{I}_1 \in \mathbb{R}^{H \times W \times C}$ and $\mathbf{I}_2 \in \mathbb{R}^{H \times W \times C}$ through the Siamese CNN branch, we can get four sets of feature maps $\{(\mathbf{F}_i^1, \mathbf{F}_i^2), i = 1, 2, 3, 4\}$ and four difference feature maps $\{\tilde{\mathbf{F}}_i, i = 1, 2, 3, 4\}$. Their sizes are $H/2^{i+1} \times W/2^{i+1} \times C_i$, where $C_1 = 64$, $C_2 = 128$, $C_3 = 256$, and $C_4 = 512$. These feature maps represent the complex local patterns and capture subtle changes between bitemporal HRRS images.

3) Siamese Transformer Branch: To capture the global context dependencies hidden in HRRS images, we introduce the deformable attention Transformer (DAT) [31] as the backbone of our Siamese Transformer branch. DAT can effectively model global information while whittling down the information loss caused by strict grids. Its primary components are the Swin-T [59] and DAT blocks. Before introducing those two blocks, we will first review the standard multihead self-attention (MSA) mechanism.

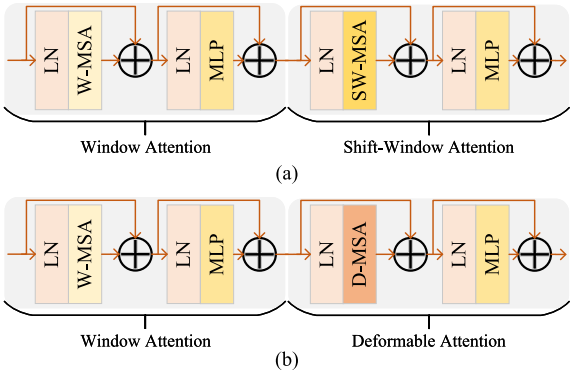


Fig. 6. Two components of the Transformer branch backbone, where "W," "SW," and "D" denote "window," "shift-window," and "deformable," respectively. (a) Swin-T block. (b) DAT block.

For a token embedding instance $\mathbf{X} \in \mathbb{R}^{n \times c}$ (where n and c indicate the number and dimension of the embeddings, respectively), the output of MSA (with h heads) is

$$\mathrm{MSA}(\mathbf{X}) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\mathbf{W}^O \tag{4}$$

where $\mathbf{W}^O \in \mathbb{R}^{c \times c}$ represents a learnable projection matrix, and $\mathrm{Concat}(\cdot)$ means the concatenation function. The definition of the ith head in MSA is

$$\begin{aligned} \mathrm{head}_i &= \mathrm{Att}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) \\ \mathbf{Q}_i &= \mathbf{X}\mathbf{W}_i^Q, \quad \mathbf{K}_i = \mathbf{X}\mathbf{W}_i^K, \ \mathbf{V}_i = \mathbf{X}\mathbf{W}_i^V \end{aligned} \tag{5}$$

where $\mathbf{Q}_i \in \mathbb{R}^{n \times c/h}$, $\mathbf{K}_i \in \mathbb{R}^{n \times c/h}$, and $\mathbf{V}_i \in \mathbb{R}^{n \times c/h}$ represent query, key, and value of $\mathbf{X} \in \mathbb{R}^{n \times c}$ in the ith head, respectively, $\mathbf{W}_i^Q$, $\mathbf{W}_i^K$, and $\mathbf{W}_i^V$ imply three learnable projection matrixes, and $\mathrm{Att}(\cdot)$ denotes the self-attention function, whose definition is

$$\mathrm{Att}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d_{\mathrm{head}}}}\right)\mathbf{V} \tag{6}$$

where $d_{\mathrm{head}} = c/h$ indicates the dimension of the self-attention head, and $\mathrm{Softmax}(\cdot)$ represents the softmax function. A common Transformer block can be developed by inserting a layer normalization (LN) before MSA and adding an LN layer and an MLP layer after MSA.

Considering the high computational costs of the common Transformer block, the Swin-T block was constructed in [59]. Its structure is shown in Fig. 6(a). A Swin-T block consists of two Transformer blocks. In the first Transformer block, a window attention strategy is introduced to replace MSA. It aims to divide the input feature map into multiple nonoverlapping windows and apply MSA to each window. We record it as W-MSA. In this fashion, the computational costs can be whittled down dramatically. In addition, to further analyze the relations between various windows, a shift-window mechanism (SW-MSA) is used to replace MSA in the second

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Transformer block. SW-MSA can be regarded as a cross-window connection scheme, and its attention function can be formulated as

$$\text{SW-Att}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d_{\text{head}}}} + \mathbf{B}\right)\mathbf{V} \tag{7}$$

where $\mathbf{B}$ denotes the relative position bias caused by the shift operations. By modeling global connectivity among different windows, SW-MSA could ensure that the contextual messages within the feature map can be fully explored.

To ensure the self-attention module could concern the important and capricious areas in the input data, the DAT block was proposed in [31], whose structure is shown in Fig. 6(b). Like the Swin-T block, there are two Transformer blocks in the DAT block. Also, the MSA mechanism in the first Transformer block is changed into W-MSA. For the second Transformer block, the MSA layer is supplanted with the deformable MSA (D-MSA). Inspired by the deformable convolution [29], D-MSA establishes an offset subnetwork to learn offsets. The subnetwork encompasses a 5×5 depth-wise convolution [64], a Gaussian error linear unit (GeLU) activation, and a 1×1 convolution. The learned offsets can guide the attention module to pay attention to the informative areas (e.g., change objects). Thus, the feature learning capacity can be improved. The attention function in D-MSA is defined as

$$\text{D-Att}\left(\mathbf{Q},\mathbf{K}',\mathbf{V}'\right) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}'^{\mathrm{T}}}{\sqrt{d_{\text{head}}}} + \mathbf{B}'\right)\mathbf{V}' \tag{8}$$

where $\mathbf{K}'$, $\mathbf{V}'$, and $\mathbf{B}'$ indicate the deformed key, value, and the relative position bias with offsets. $\mathbf{K}'$ and $\mathbf{V}'$ can be obtained by

$$\begin{aligned} &\mathbf{K}' = \mathbf{X}'\mathbf{W}^{K}, \quad \mathbf{V}' = \mathbf{X}'\mathbf{W}^{V} \\ &\mathbf{X}' = \phi(\mathbf{X}; p + \Delta p), \quad \Delta p = \delta_{\text{offset}}(\mathbf{Q}) \end{aligned} \tag{9}$$

where $\mathbf{X}'$ implies deformed input instance $\mathbf{X}$, $\phi(\cdot)$ denotes the bilinear interpolation function, Δp expresses the learnable offsets, $\delta_{\text{offset}}(\cdot)$ represents the function of the offset subnetwork, and $\mathbf{Q} = \mathbf{X}\mathbf{W}^{Q}$ indicates the query.

As exhibited in Fig. 3, our Siamese Transformer branch incorporates two identical DATs, each of which has four stages. The two Swin-T blocks are located in the first two stages, while the DAT blocks are embedded in the last two stages. Besides, each stage contains a patch embedding layer for converting the feature map to tokens and downsampling. Note that there are three DAT blocks in Stage 3. Four DEMs connect two DATs to produce the difference feature maps. In sum, when the bitemporal HRRS images $\mathbf{I}_1 \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{I}_2 \in \mathbb{R}^{H\times W\times C}$ are input to the Siamese Transformer branch, both the multilevel feature maps $\{(\mathbf{G}_i^1, \mathbf{G}_i^2), i = 1, 2, 3, 4\}$ and their corresponding difference feature maps $\{\tilde{\mathbf{G}}_i, i = 1, 2, 3, 4\}$ can be achieved. Their dimensions are the same as the outputs of the Siamese CNN branch. These features carry multiscale global information, which is helpful for HRRS image CD.

C. Fusion Decoder

To generate accurate change maps, we develop the fusion decoder in this section. As illustrated in Fig. 3, our fusion decoder consists of four CTFMs (corresponding to four stages in the dual-branch encoder) and a simple classifier. Their functions are to fuse local and global features from the Siamese CNN and Siamese Transformer branches and to produce change maps.

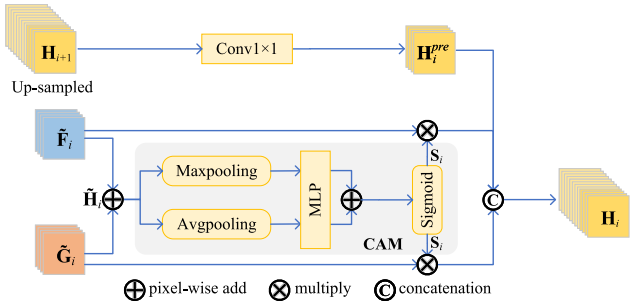


Fig. 7. Structure of CTFM, which involves pixel-wise add, multiply, concatenation, max pooling, average pooling, 1×1 convolution, and Sigmoid operations.

Four CTFMs have the same structure (see Fig. 7). For clarity, we use the CTFM located in the third stage as an example to explain it in detail. For $\tilde{\mathbf{F}}_3 \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$ and $\tilde{\mathbf{G}}_3 \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$, CTFM first adds them together to integrate the fine local information and global context dependencies coarsely, and we record the summation feature as $\tilde{\mathbf{H}}_3 \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$. Then, $\tilde{\mathbf{H}}_3$ would be fed into a CAM [18] to obtain the channel-wise weight $\mathbf{S}_3$, which contains a max pooling layer, an average pooling layer, an MLP (constructed by two 1×1 convolutions), and a sigmoid function. Next, CTFM would multiply $\mathbf{S}_3$ to $\tilde{\mathbf{F}}_3$ and $\tilde{\mathbf{G}}_3$ individually to highlight the important messages from the channel aspects. Finally, the product results will be concatenated with feature $\mathbf{H}_3^{\text{pre}} \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$ to generate the fused feature map $\mathbf{H}_3 \in \mathbb{R}^{(H/16)\times(W/16)\times 3C_3}$. Here, $\mathbf{H}_3^{\text{pre}}$ is achieved by the upsampling operation and 1×1 convolution based on $\mathbf{H}_4 \in \mathbb{R}^{(H/32)\times(W/32)\times 2C_4}$. The concatenation can not only combine weighted local and global features, but also aggregate upper-level features to complete interscale information fusion. The above process can be generalized by the following formulas:

$$\begin{aligned} &\mathbf{H}_i = \text{Concat}\left(\mathbf{S}_i\tilde{\mathbf{F}}_i, \mathbf{S}_i\tilde{\mathbf{G}}_i, \mathbf{H}_i^{\text{pre}}\right) \\ &\mathbf{S}_i = \sigma\left(\text{MLP}\left(\text{Maxpool}\left(\tilde{\mathbf{H}}_i\right)\right) + \text{MLP}\left(\text{Avepool}\left(\tilde{\mathbf{H}}_i\right)\right)\right) \\ &\tilde{\mathbf{H}}_i = \tilde{\mathbf{F}}_i + \tilde{\mathbf{G}}_i, \quad \mathbf{H}_i^{\text{pre}} = f_{\text{Conv1}\times1}(\text{Up}(\mathbf{H}_{i+1})) \end{aligned} \tag{10}$$

where $\text{Concat}(\cdot)$ indicates the concatenation operation at the channel-wise, $\text{Maxpool}(\cdot)$ and $\text{Avepool}(\cdot)$ mean the max and average pooling layer, respectively, $\text{MLP}(\cdot)$ denotes the MLP function, $\sigma(\cdot)$ implies the sigmoid function, $\text{Up}(\cdot)$ expresses the upsampling, and $f_{\text{Conv1}\times1}(\cdot)$ specifies the 1×1 convolution. Note that there is no $\mathbf{H}_i^{\text{pre}}$ in the concatenation process in Stage 4. Also, the upsampling in the decoder is accomplished by the bilinear interpolation algorithm.

At the top of the encoder, a simple classifier is developed to generate change maps. Specifically, the classifier accommodates two 3×3 convolutions and a softmax function. In addition, a BN layer and a ReLU activation are added after the first convolution layer. When the feature map $\mathbf{H}^* \in \mathbb{R}^{H\times W\times 3C_1}$ is fed into the classifier, the change probability prediction $\mathbf{P} \in \mathbb{R}^{H\times W\times 2}$ can be obtained, where $\mathbf{H}^*$ is

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upsampled by $\mathbf{H}_1$. Finally, the binary change map $\mathbf{M} \in \mathbb{R}^{H\times W}$ is generated by argmax operation on the prediction $\mathbf{P}$, where $\mathbf{M}(i, j) = 1$ indicates the changed pixels and $\mathbf{M}(i, j) = 0$ implies the unchanged pixels.

Generally, the sample imbalance problem exists in CD tasks, where the number of unchanged pixels exceeds that of changed pixels. To mitigate this problem, we combine the cross-entropy loss $\mathcal{L}_{\text{ce}}$ and dice loss $\mathcal{L}_{\text{dice}}$ to train our WNet. For the predicted binary change map $\mathbf{M}$ and the ground-truth $\mathbf{Y}$, the definitions of $\mathcal{L}_{\text{ce}}$ and $\mathcal{L}_{\text{dice}}$ can be expressed as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\mathbf{Y}_i \log \mathbf{M}_i + (1-\mathbf{Y}_i)\log(1-\mathbf{M}_i)\right]$$
$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_{i=1}^{N}\mathbf{M}_i\mathbf{Y}_i}{\sum_{i=1}^{N}(\mathbf{M}_i+\mathbf{Y}_i)+\epsilon} \tag{11}$$

where $\epsilon = 1e-7$ is set to avoid being divided by zero and N denotes the number of pixels. In the training process, $\mathcal{L}_{\text{ce}}$ treats pixels equally and $\mathcal{L}_{\text{dice}}$ pays attention to the changed pixels. Therefore, the combination of $\mathcal{L}_{\text{ce}}$ and $\mathcal{L}_{\text{dice}}$ can balance unchanged and changed pixels to a certain extent. Finally, the total loss function $\mathcal{L}$ for our model can be formulated as

$$\mathcal{L} = \lambda\mathcal{L}_{\text{ce}} + (1-\lambda)\mathcal{L}_{\text{dice}} \tag{12}$$

where λ is a parameter that controls the specific gravity of two loss functions.

IV. Experiment

A. Datasets Introduction

To evaluate the proposed WNet, the following four HRRS image CD datasets are selected in this article.

1) Season-Varying CD Dataset (SVCD) [65]: SVCD comprises two sets of season-varying HRRS image pairs collected from Google Earth. In the first set, there are seven image pairs with the size of 4725 × 2700. Their changes reach the demands of manual annotation. In the second set, there are four image pairs with the size of 1900 × 1000. The changes within them are minimal. Thus, the objects with various scales are added by researchers to synthesize the diverse changes. The spatial resolutions of the above 11 image pairs are varied from 3 to 100 cm per pixel. After the preprocessing (cropping and rotation), the final training/validation/testing sets enclose 10 000/2998/3000 pairs with the size of 256 × 256.

2) WuHan University Dataset (WHU-CD) [66]: WHU-CD contains a pair of aerial images with a size of 32 507 × 15 354. Their resolutions are 0.2 m per pixel. The main changes in this dataset are buildings. Since this dataset has no unified split rule, we cut the original aerial images into patches with a size of 256 × 256 pixels nonoverlapped and randomly divided them into 5947/743/744 patch pairs for training/validation/testing, respectively.

3) LEarning, Vision, and RS (LEVIR-CD) [14]: LEVIR-CD contains 637 image pairs (obtained from Google Earth) with a size of 1024 × 1024, whose spatial resolutions are 0.5 m. These images cover over 20 regions and encompass 31 333 change instances, concentrating on the buildings. According to the original literature, we crop these images into 256 × 256 patches and construct the training/validation/testing sets involving 7120/1024/2048 patch pairs.

4) Sun Yat-Sen University Dataset (SYSU-CD) [8]: SYSU-CD is a large-scale CD dataset, which consists of 20 000 256 × 256 HRRS image pairs, whose spatial resolutions are 0.5 m. Notably, there are multiple change targets in the SYSU-CD dataset, such as urban buildings, vegetation, and maritime vessels. This increases the difficulty of CD. Consistent with the common division strategy, 12 000/4000/4000 pairs are selected to establish the training/validation/testing set, respectively.

B. Experimental Settings

All experiments are conducted on a single NVIDIA Tesla V100 with 32G video memory, and the compilation platform is PyTorch. The common data augmentation schemes (e.g., horizontal/vertical flipping, scale cropping, and blurring) are used in the training phase to ensure the WNet's stability. The dual-branch backbones of WNet are initialized by the pretrained parameters of ResNet18 [33] and DAT-tiny [31] using ImageNet. Other parts of our WNet are initialized randomly. Here, the Adam optimizer with the parameters $\beta_1 = 0.5$ and $\beta_2 = 0.999$ is adopted to complete the training. The training epochs, batch size, and learning rate equal 250, 16, and 1e-4, respectively. The learning rate for the first half of epochs remains and then would be linearly decayed. One free parameter should be set in advance, that is, the loss control parameter λ [see (12)]. In the following experiments, we fix $\lambda = 0.9$ unless otherwise stated, and its influence will be discussed in Section IV-E.

To investigate WNet's performance clearly, we select five assessment indicators [1], [14]: precision (Pre), recall (Rec), intersection-of-union (IoU), F1 score, and overall accuracy (OA). Their definitions are

$$\text{Pre} = \frac{\text{TP}}{\text{TP}+\text{FP}},\quad \text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}},\quad \text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}}$$
$$\text{F1} = \frac{2\times\text{Pre}\times\text{Rec}}{\text{Pre}+\text{Rec}},\quad \text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{13}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative pixels, respectively.

C. Performance of WNet

We select eight methods for comparison to evaluate the performance of the proposed WNet, including FC-Siam-conc [37], FC-Siam-diff [37], DSAMNet [8], SNUNet [43], ISNet [49], BIT [60], SwinSUNet [56], and ICIF-Net [61]. Here, the last three are Transformer-based methods, and the rest are CNN-based approaches. All compared methods are completed using the codes released by the authors. Their free parameters are set according to the original literature, and the experimental settings are the same as mentioned in Section IV-B.

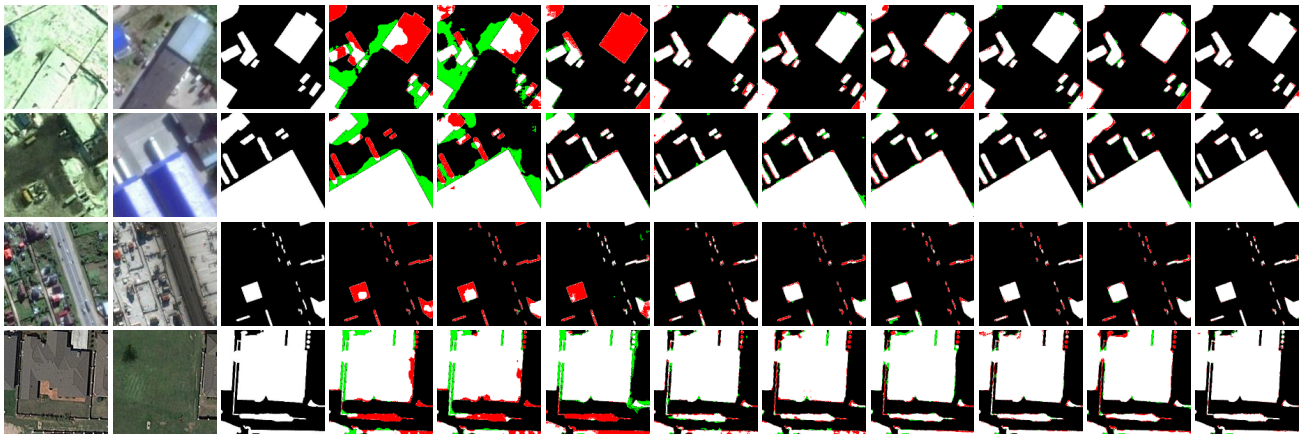
1) Numerical Results: The quantitative results of different models counted on four datasets are shown in Table I, where the best values are in bold and the second-best values are underlined. We can easily find that the proposed WNet performs best, and there are distinct performance gaps between

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TABLE I

PERFORMANCE OF DIFFERENT MODELS COUNTED ON FOUR DATASETS. FOR PRE, REC, IOU, F1, AND OA VALUES (%), THE BEST VALUES ARE IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Models	SVCD					WHU-CD					LEVIR-CD					SYSU-CD				
	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA
FC-Siam-conc [37]	83.92	69.83	61.59	76.23	94.86	45.11	76.63	39.65	56.79	95.37	86.95	84.84	75.26	85.88	98.58	82.17	72.06	62.32	76.78	89.72
FC-Siam-diff [37]	86.34	66.87	60.47	75.36	94.84	44.76	87.21	42.00	59.16	95.22	88.53	78.28	71.07	83.09	98.38	72.69	79.46	61.20	75.93	88.12
DSAMNet [8]	93.41	91.94	86.35	92.67	98.26	81.91	89.58	74.78	85.57	98.90	87.43	83.04	74.18	85.18	98.45	**83.86**	73.32	64.25	78.24	89.00
SNUNet [43]	95.70	95.08	91.19	95.39	98.92	86.31	90.12	78.85	88.17	99.08	91.43	89.36	82.46	90.38	99.03	83.46	73.92	64.48	78.40	90.40
ISNet [49]	95.18	94.43	90.12	94.80	98.78	87.84	87.27	77.87	87.56	99.01	90.88	89.31	81.97	90.09	99.00	81.87	74.15	63.69	77.82	90.03
BIT [60]	95.17	94.57	90.24	94.87	98.79	88.69	88.28	79.35	88.49	99.09	**92.25**	88.32	82.22	90.24	99.03	79.77	74.84	62.90	77.22	89.60
SwinSUNet [56]	95.96	94.58	90.96	95.26	98.89	90.35	87.41	79.94	88.85	99.12	90.84	89.35	81.97	90.09	99.00	81.99	73.30	63.13	77.40	89.90
ICIFNet [61]	95.34	94.76	90.56	95.05	98.83	90.91	89.65	82.28	90.27	99.23	91.13	89.28	82.14	90.19	99.01	80.30	74.97	63.32	77.54	89.76
WNet(ours)	**97.40**	**97.71**	**95.23**	**97.56**	**99.42**	**92.37**	**90.15**	**83.91**	**91.25**	**99.31**	91.16	**90.18**	**82.93**	**90.67**	**99.06**	81.71	**79.58**	**67.55**	**80.64**	**90.98**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 8. Different change maps of four bitemporal HRRS image pairs that are randomly selected from the SVCD dataset. Here, white, black, green, and red denote TP, TN, FP, and FN pixels, respectively. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-Siam-conc. (e) FC-Siam-diff. (f) DSAMNet. (g) SNUNet. (h) ISNet. (i) BIT. (j) SwinSUNet. (k) ICIFNet. (l) WNet (ours).

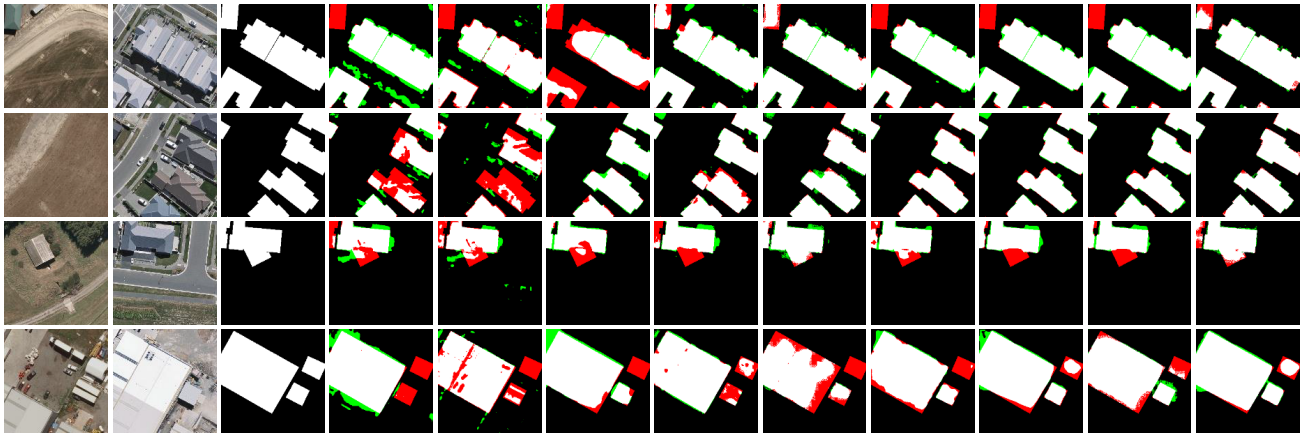
our WNet and the second-best models in most cases. Taking SVCD as an example, the improvements achieved by WNet in F1 scores are 21.33% (over FC-Siam-conc), 22.20% (over FC-Siam-diff), 4.89% (over DSAMNet), 2.17% (over SNUNet), 2.76% (over ISNet), 2.69% (over BIT), 2.30% (over SwinSUNet), and 2.51% (over ICIF-Net). Taking WHU-CD as another example, the enhancements achieved by WNet in IoU are 44.26% (over FC-Siam-conc), 41.91% (over FC-Siam-diff), 9.13% (over DSAMNet), 5.06% (over SNUNet), 6.04% (over ISNet), 4.56% (over BIT), 3.97% (over SwinSUNet), and 1.63% (over ICIF-Net). An encouraging investigation is that the behavior of WNet is stable for different datasets. For instance, SNUNet's performance is suitable for SVCD, LEVIR-CD, and SYSU-CD. However, its behavior for WHU-CD is not as good as expected. For another instance, ICIF-Net performs well for WHU-CD. However, its behavior is relatively weaker for the other three datasets. Compared to other methods, the proposed WNet can achieve satisfactory results for all datasets, which demonstrates the generalization ability of our model.

The above observations illustrate the effectiveness of WNet from various aspects. The reasons for these positive results are threefold. First, both CNN and Transformer's advantages are leveraged in WNet, which is beneficial for exploring the local and global information hidden in HRRS images. Second, the developed DEM can highlight the change areas between bitemporal images. Along with the multiscale scheme, sufficient and diverse changed clues can be mined for the final CD. Third, the designed CTFM can fully fuse the local knowledge and global contexts, which ensures the quality of change maps. Although WNet's behavior is assertive, we cannot neglect that its performance in Pre is not optimal for LEVIR-CD and SYSU-CD. This is because the embedded deformable ideas make WNet focuses on the changing objects of interest, resulting in the final change maps having relatively more FP and fewer FN pixels. Fortunately, our Pre values are still in a reasonable range.

2) Visual Results: The examples of change maps achieved by different methods are displayed in Figs. 8–11, where white, black, green, and red represent TP, TN, FP, and FN pixels, respectively. By investigating these visual results, we can get the following conclusions.

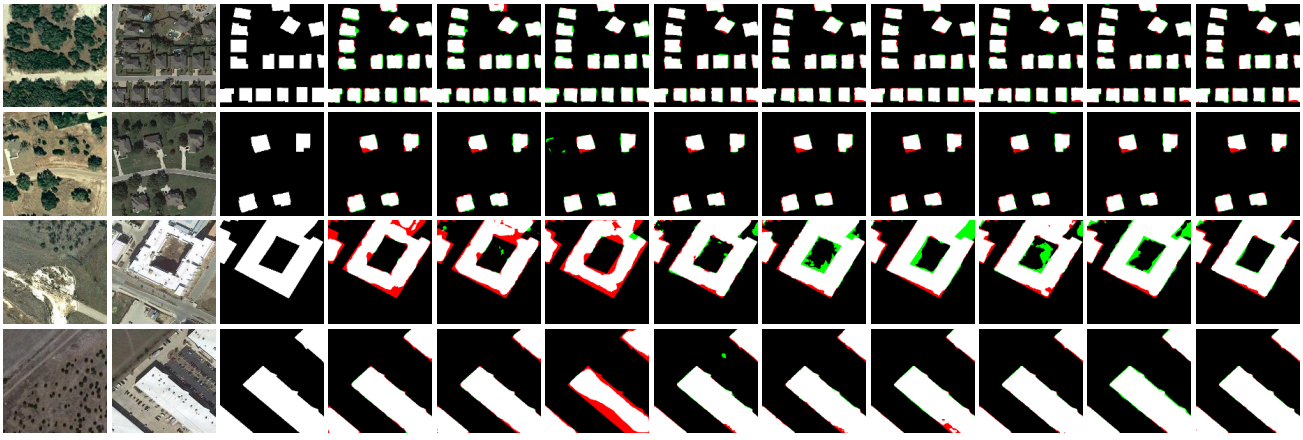
1) WNet can effectively mitigate the negative impacts of various environmental conditions on CD. First, as shown in the first and third rows of Fig. 8, WNet can detect changes of interest under seasonal variation, such as white buildings that are similar in color to snow. Instead, the detection results of other methods, such as DSAMNet, are not good enough (see red areas). Second, in the first and fourth rows of Fig. 9, WNet can distinguish the buildings with a similar color appearance, while other methods are missing these areas. Third, WNet can

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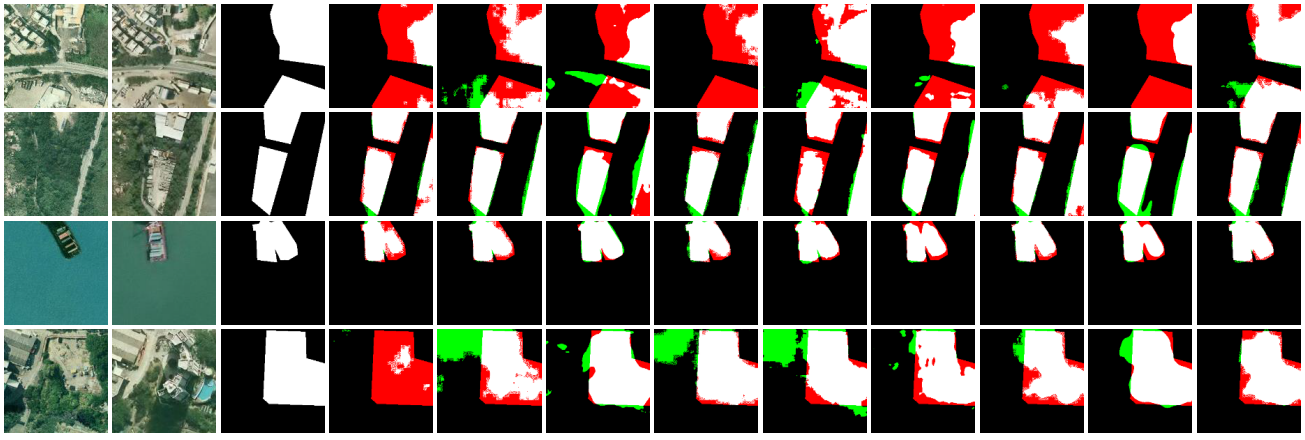
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 9. Different change maps of four bitemporal HRRS image pairs that are randomly selected from the WHU-CD dataset. Here, white, black, green, and red denote TP, TN, FP, and FN pixels, respectively. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-Siam-conc. (e) FC-Siam-diff. (f) DSAMNet. (g) SNUNet. (h) ISNet. (i) BIT. (j) SwinSUNet. (k) ICIFNet. (l) WNet (ours).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 10. Different change maps of four bitemporal HRRS image pairs that are randomly selected from the LEVIR-CD dataset. Here, white, black, green, and red denote TP, TN, FP, and FN pixels, respectively. (a) T1 image. (b) T2 image. (c) ground truth. (d) FC-Siam-conc. (e) FC-Siam-diff. (f) DSAMNet. (g) SNUNet. (h) ISNet. (i) BIT. (j) SwinSUNet. (k) ICIFNet. (l) WNet (ours).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 11. Different change maps of four bitemporal HRRS image pairs that are randomly selected from the SYSU-CD dataset. Here, white, black, green, and red denote TP, TN, FP, and FN pixels, respectively. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-Siam-conc. (e) FC-Siam-diff. (f) DSAMNet. (g) SNUNet. (h) ISNet. (i) BIT. (j) SwinSUNet. (k) ICIFNet. (l) WNet (ours).

generate high-quality change maps (which are close to the ground truth) under different lighting conditions (see the third line of Fig. 10). On the contrary, the change maps of other methods all have a large number of FP or FN pixels.

2) WNet can effectively identify both large-scale and fine-grained changes with different resolutions. As illustrated in the third and fourth lines of Fig. 8(l), apart from detecting large building structures, the changes in small targets (such as vehicles and trees) can also be detected

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TABLE II
COMPARISON OF MODEL EFFICIENCY WITH THE PARAMETER SIZE (PARAM) AND FLOPS

Models	Param (M)	FLOPs (G)
FC-Siam-conc [37]	1.55	5.33
FC-Siam-diff [37]	1.35	4.73
DSAMNet [8]	16.95	75.29
SNUNet [43]	10.20	44.38
ISNet [49]	14.36	9.58
BIT [60]	3.50	10.63
SwinSUNet [56]	40.95	11.19
ICIFNet [61]	23.84	25.41
WNet(ours)	43.07	19.20

by our WNet. Other methods perform unsatisfactorily in small targets, resulting in many FN pixels (in red). Similarly, the fourth line of Fig. 9 also demonstrates that WNet can deal with both large and small changes. In this pair of examples, only WNet has detected the disappearance of the white shed in the T1 image.

3) WNet can effectively detect changes in complex geographical environments. SYSU-CD is a dataset with multiple complex targets and backgrounds, including buildings, trees, ship targets, land, vegetation, and ocean (see Fig. 11). Benefiting from the Siamese CNN and Siamese Transformer branches, both the detailed local information and the global context knowledge can be mined by WNet. Thus, compared to other models, WNet can produce better change maps with fewer red and green pixels.

3) Model Complexity: Apart from numerical and visual results, we also count the parameter size (Param) and floating-point operations (FLOPs) of WNet and other models. Note that the input data sizes are $3 \times 256 \times 256$ when we estimate FLOPs. The results are displayed in Table II. By analyzing them, we can find that the parameter sizes of the Transformer-based models are larger than that of the CNN-based methods due to the self-attention mechanism in the Transformer. Furthermore, WNet is the heaviest one among all models. This is because there are two dual-branch networks in its encoder, which leads to a large number of parameters to be trained. Fortunately, owing to the simplicity of structure and module design, WNet's FLOPs are lower than some complex methods (e.g., DSAMNet, SNUNet, and ICIF-Net). In sum, our model's parameter size and FLOPs stay in an acceptable range. Developing the lightweight version is one of our further works.

D. Ablation Studies

This section studies each part of WNet from the following four aspects. First, the contributions of the proposed D-Residual block are discussed. Then, we investigate if the dual-Siamese structure helps CD or not. Next, the rationality of choosing DAT as the backbone of our Siamese Transformer is inspected. Finally, the effectiveness of the proposed DEM and CTFM will be studied. Here, we only count the IoU and F1 values of different models for clarity. Also, experimental settings and free parameters are set according to the contents mentioned in Section IV-B. Note that the bitemporal difference feature generation (in the encoding stage) and dual-branch feature fusion (in the decoding stage) are completed through the channel dimension concatenation for the first three scenarios.

TABLE III
ABLATION STUDIES OF THE DUAL-SIAMESE BRANCH STRUCTURE AND ITS COMPONENTS. THE BEST IOU AND F1 VALUES (%) ARE IN BOLD

Networks	SVCD		WHU-CD		LEVIR-CD		SYSU-CD	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1
(a) Effectiveness of D-Residual Block								
Net-0	90.42	94.97	72.28	83.91	79.65	88.67	62.40	76.85
Net-1	90.62	95.08	75.44	86.00	81.04	89.53	62.65	77.04
(b) Effectiveness of Dual Siamese Branch								
Net-1	90.62	95.08	75.44	86.00	81.04	89.53	62.65	77.04
Net-2	93.83	96.82	80.51	89.20	80.21	89.02	62.32	76.79
Net-3	94.62	97.24	82.16	90.21	81.30	89.69	62.96	77.26
(c) Influence of Different Transformers								
Net-3	94.62	97.24	82.16	90.21	81.30	89.69	62.96	77.26
Net-4	94.23	97.03	81.80	89.99	80.76	89.36	62.69	77.07
Net-5	94.20	97.01	80.90	89.44	80.43	89.15	62.82	77.17
Net-6	92.65	96.19	77.09	87.07	81.20	89.63	61.11	75.86
Net-7	94.28	97.06	79.11	88.34	81.11	89.57	62.20	76.70
Net-8	93.46	96.62	77.26	87.17	81.27	89.67	61.47	76.14
(d) Effectiveness of Different Modules								
Net-3	94.62	97.24	82.16	90.21	81.30	89.69	62.96	77.26
Net-9	94.95	97.41	83.45	90.98	82.17	90.22	66.91	80.18
Net-10	**95.23**	**97.56**	**83.91**	**91.25**	**82.93**	**90.67**	**67.55**	**80.64**
(e) Effectiveness of Integration of CNN and Transformer								
Net-10	**95.23**	**97.56**	**83.91**	**91.25**	**82.93**	**90.67**	**67.55**	**80.64**
Net-11	93.15	96.46	78.62	88.03	82.45	90.38	66.13	79.61
Net-12	94.21	97.02	80.96	89.48	81.29	89.67	63.35	77.57

1) Effectiveness of D-Residual Block: The D-Residual block aims to alleviate the limitations of exploring HRRS images with regular convolutional kernels by introducing deformable convolution. To demonstrate its effectiveness, we construct two simple HRRS image CD networks.

Net-0: Only Siamese CNN branch (Residual block).
Net-1: Only Siamese CNN branch (D-Residual block).

Net-0 means there is a Siamese network with two normal ResNet18 [33]. Net-1 indicates that the normal residual blocks within ResNet18 are replaced by the D-Residual blocks. As mentioned at the beginning of this section, the difference feature maps and feature fusion results are produced by the channel-wise concatenation. The IoU and F1 values of the two networks are shown in Table III(a). Obviously, Net-1 outperforms Net-0 in all cases. This illustrates that the D-Residual block is helpful in mining valuable information from HRRS images for CD tasks.

2) Effectiveness of Dual-Siamese Branch: As mentioned in Section III-B, the dual-Siamese branch encoder plays an

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essential role in our WNet. To verify if the dual-Siamese structure performs better than the single Siamese branch structure, we design the following three networks.

Net-1: Only Siamese CNN branch (D-residual block).
Net-2: Only Siamese transformer branch (DAT).
Net-3: Dual-Siamese branch (Net-1 + Net-2).

Net-1 and Net-2 imply that the CD network only contains a Siamese CNN encoder or a Siamese Transformer encoder. Net-3 denotes that the encoder involves a Siamese CNN and a Siamese Transformer in parallel. Similar to the first scenario, the channel-wise concatenation also completes the difference feature generation and different branch feature fusion steps. The IoU and F1 values of three networks counted on four datasets are exhibited in Table III(b).

From the observation of numerical results, we can find that the performance of Net-2 is better than that of Net-1 in the SVCD and WHU-CD datasets, and the behavior of Net-1 is stronger than that of Net-2 in the LEVIR-CD and SYSU-CD datasets. Also, the performance gaps between Net-1 and Net-2 in SVCD and WHU-CD are higher than that in LEVIR-CD and SYSU-CD. These observations imply that the Transformer is more suitable for the current HRRS image CD datasets. When we combine them (i.e., Net-3), the CD results counted on different datasets are improved to a large degree. This illustrates that the dual-Siamese branch structure is conducive to CD tasks since local and global information can be explored for generating change maps.

3) Influence of Different Transformers: As discussed in Section III-B3, DAT [31] is the backbone of our Siamese Transformer branch. To study whether this choice is suitable, we select five other popular Transformers and construct five new networks.

Net-4: Dual-Siamese branch (Net-1 + Swin-T).
Net-5: Dual-Siamese branch (Net-1 + PVT).
Net-6: Dual-Siamese branch (Net-1 + SegFormer).
Net-7: Dual-Siamese branch (Net-1 + MPViT).
Net-8: Dual-Siamese branch (Net-1 + BiFormer).

Net-4, Net-5, Net-6, Net-7, and Net-8 indicate that we use Swin-T [59], PVT [67], SegFormer [52], MPViT [53], and BiFormer [54] to replace DAT in Net-3. These Transformers are all operated downstream based on strictly grid-partitioned patches. Here, the Transformers are initialized by the pretrained parameters of Swin-tiny [59], PVTv2-B1 [67], SegFormer-B1 [52], MPViT-tiny [53], and BiFormer-tiny [54]. These five networks' performance is summarized in Table III(c), and Net-3's results are also exhibited for reference. By analyzing the results, it is easy for us to find that Net-3 (embedded by DAT) achieves better performance than all other Nets for all datasets. The reason behind this is that DAT can effectively alleviate information loss caused by strict grid division and improve CD accuracy. This observation confirms that choosing DAT as the backbone of the Siamese Transformer branch is suitable.

4) Effectiveness of Different Modules: After studying the D-Residual block, dual-Siamese structure, and DAT's functions, the other two important components of WNet (i.e., DEM and CTFM) will be discussed here. DEM is used to generate difference feature maps and enhance change information, while CTFM is used to fuse local, global, and cross-scale knowledge. To certify the usefulness of DEM and CTFM, we construct the following two networks based on Net-3.

Net-9: Dual-Siamese branch + DEM (Net-3 + DEM).
Net-10: Proposed WNet (Net-9 + CTFM).

Net-9 means that we add DEM into Net-3 to ensure the quality of generated difference feature maps. Net-10 is actually our WNet. These two networks and Net-3's performance counted on four HRRS image CD datasets are shown in Table III(d).

Compared with Net-3, the improvements achieved by Net-9 in IoU/F1 scores are 0.33%/0.17% (SVCD), 1.29%/0.77% (WHU-CD), 0.87%/0.53% (LEVIR-CD), and 3.95%/2.92% (SYSU-CD). Compared with Net-9, the enhancements obtained by Net-10 in two assessment criteria are 0.28%/0.15% (SVCD), 0.46%/0.27% (WHU-CD), 0.76%/0.45% (LEVIR-CD), and 0.64%/0.46% (SYSU-CD). The above findings illustrate that DEM and CTFM play positive roles in our WNet.

5) Effectiveness of Integration of CNNs and Transformers: As mentioned in Section I, one of the motivations of this work is to integrate the advantages of CNNs and Transformers. To confirm whether this integration is proper, we construct the following networks.

Net-11: Two Siamese CNN branches + DEM + CTFM.
Net-12: Two Siamese Transformer branches + DEM + CTFM.

Net-11 and Net-12 denote the dual-Siamese branch encoder composed of two Siamese CNNs and two Siamese Transformers, respectively. All of them are embedded with DEMs. Besides, the fusion decoders consisting of fusion modules are added after the encoders. Their IoU and F1 values counted on four datasets are exhibited in Table III(e), and the results of our WNet (i.e., Net-10) are also exhibited for reference.

By observing the results, we can easily find that Net-10 outperforms the other two nets in all cases. Taking SVCD as an example, the enhancements achieved by Net-10 in IoU value are 2.08% (Net-11) and 1.02% (Net-12). Taking SYSU-CD as another instance, the improvements obtained by Net-10 in F1 value are 1.03% (Net-11) and 3.07% (Net-12). These positive results demonstrate that the combination of CNNs and Transformers is conducive to finding the change information hidden in bitemporal HRRS images.

E. Paramter Analysis

There is a free parameter in our WNet, that is, the weight λ controlling the contributions of the two terms of (12). In this section, we will study its influence on our model. To this end, we change its value from 0.1 to 1.0 with an interval of 0.1. Then, various WNets can be obtained based on different λ, and their F1 values counted on four datasets are exhibited in Fig. 12. By observing these figures, we can find the following points. First, different WNets' behavior becomes stronger when λ values are increased from 0.1 to 0.9. Also, the peak values appear at $\lambda = 0.9$ coincidentally. This implies that the cross-entropy loss $\mathcal{L}_{ce}$ is relatively more important.

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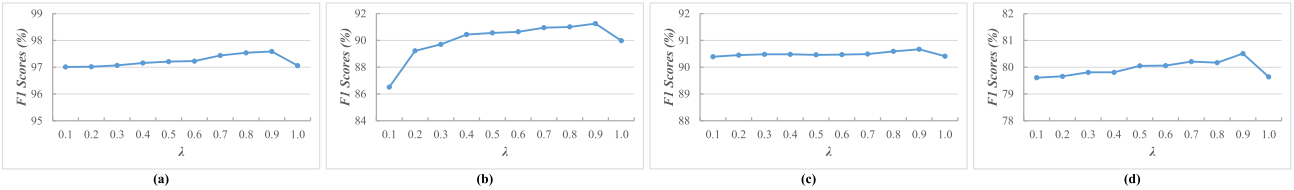


Fig. 12. F1 scores of different WNets based on different λ, which are counted on four datasets. (a) SVCD. (b) WHU-CD. (c) LEVIR-CD. (d) SYSU-CD.

However, the performance of WNets is whittled down distinctly when $\lambda = 1.0$, which points out that the dice loss $\mathcal{L}_{\text{dice}}$ is indispensable. Second, the performance of different WNets is stable for SVCD and LEVIR-CD. Nevertheless, for WHU-CD and SYSU-CD, their behavior changes are apparent. This is mainly because the distributions of the change instances are different. Fortunately, all behavioral changes are within a reasonable range. Based on the above discussion, the optimal value of λ is 0.9. Also, we suggest to the readers that they can tune this parameter around 0.9 when they apply our WNet to new data.

V. Conclusion

This article proposes a dual-Siamese branch hierarchical network (i.e., WNet) for HRRS image CD. WNet is composed of a dual-Siamese branch encoder and a fusion decoder. The encoder encloses a Siamese CNN and Siamese Transformer in parallel. Each has four stages, which aim to extract diverse local information and global long-range context dependencies from bitemporal HRRS images. We also introduce deformable convolution and attention into the Siamese CNN and Siamese Transformer to enhance the usefulness of the obtained knowledge. In this fashion, WNet will have more flexible receptive fields to understand the diverse spatial areas within HRRS images so that the information quantity of the generated features can be ensured. Meanwhile, the simple, yet effective DEM is developed to highlight interest changes. Through straightforward pixel-wise subtraction and channel dimension concatenation, DEM can emphasize the changes of interest and suppress the irrelevant changes in a learnable manner. After adding DEMs to different stages of WNet's encoder, the discriminative difference feature maps can be produced. The fusion decoder consists of CTFMs and a classifier. CTFM is developed based on CAM, focusing on fusing local, global, and cross-level clues at different stages. The classifier is constructed to predict change maps, which encompasses two convolution layers and a softmax function. The encouraging numerical and visual results counted on four public HRRS image CD datasets demonstrate that our WNet performs better than many existing methods. Although WNet's behavior is satisfactory, its parameter size is relatively large due to the dual-Siamese branch structure. In the future, we will pay attention to lightening WNet while maintaining its performance.

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